UIC UNIVERSITY OF ILLINOIS AT CHICAGO

Title:	Revision of the Bachelor of Science in Chemical Engineering
Sponsor:	*Department of Chemical Engineering*
Executive Summary:	The department is proposing to add a new technical elective course, CHE 451 Renewable Energy Technologies to the chemical engineering program. The course was developed under the general CHE 494 “Selective Topics in Chemical Engineering” rubric and has been offered Fall semester since 2011 It will be offered to undergraduate students as a 3 credit hour course and to graduate students as a 4 credit hour course and will give the students another course option to fulfill their technical elective requirement.
Description:	Addition of one technical elective inside the major rubric; new course (CHE 451). This change does not affect the total credit hours required for the program.
Justification:	The course provides senior undergraduate and graduate students fundamentals of renewable energy sources, energy storage and conversion technologies, and technical and economic analysis of the promises and challenges of the Hydrogen Economy. The course has been well-received by undergraduate and graduate chemical engineering students and satisfies the technical elective requirement for the bachelor’s degree. The topics covered in the course also integrate holistically with the overall Chemical Engineering curriculum and undergraduate student learning experience.
Catalog Statement:	*See below*
Minority Impact Statement:	*The changes will clarify the degree requirements for all students and will not adversely impact minority students.*
Budgetary and Staff Implications:	*None expected*
Library Resource Implications:	*None expected*
Space Implications:	*None expected*

Unit (e.g. department) approval date: *College (educational policy committee, faculty) approval dates:*	Provide unit dates, including department, college/school committees, faculty Approvals, as applicable. Chemical engineering faculty approval: 09/23/16 Chemical Engineering department approval: Education Policy Committee approval:
Contact Person:	Alan Zdunek zdunek@uic.edu
Proposed Effective Date/Term:	Fall, 2017

Current	Revised
BS in Chemical Engineering	BS in Chemical Engineering
Degree Requirements To earn a Bachelor of Science in Chemical Engineering degree from UIC, students need to complete University, college, and department degree requirements. The Department of Chemical Engineering degree requirements are outlined below. Students should consult the *College of Engineering* section for additional degree requirements and college academic policies.	SAME
BS in Chemical Engineering Degree Requirements Hours Nonengineering and General Education Requirements 73 Required in the College of Engineering 49 Technical Elective[a] 3 Electives outside the Major Rubric[a] 3 **Total Hours—BS in Chemical Engineering 128** a *Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.*	SAME
Nonengineering and General Education Requirements ENGL 160— Academic Writing I: Writing for Academic and Public Contexts 3 ENGL 161— Academic Writing II: Writing for Inquiry and Research 3 Exploring World Cultures course[a] 3 Understanding the Creative Arts course[a] 3 Understanding the Past course[a] 3 Understanding the Individual and Society course[a] 3 Understanding U.S. Society course[a] 3 MATH 180—Calculus I[b] 4 MATH 181—Calculus II[b] **4** MATH 210—Calculus III[b] 3 MATH 220—Introduction to Differential Equations 3 PHYS 141—General Physics I (Mechanics)[b] 4	SAME

PHYS 142—General Physics II (Electricity and Magnetism)[b] 4 CHEM 122- General College Chemistry I[b] (Lecture) 4 CHEM 123- General College Chemistry I[b](Laboratory) 1 <u>OR</u> CHEM 116 – Honors and Major General and Analytical Chemistry I 5 CHEM 124—General College Chemistry II[b](Lecture) 4 CHEM 125—General College Chemistry II[b](Lab) 1 <u>OR</u> CHEM 118 – Honors and Major General and Analytical Chemistry II 5 CHEM 222—Analytical Chemistry[c] 4 CHEM 232—Organic Chemistry I 4 CHEM 233- Organic Chemistry Laboratory 2 CHEM 234—Organic Chemistry II 4 CHEM 342—Physical Chemistry I 3 CHEM 346—Physical Chemistry II 3 Total Hours- Nonengineering and General Education Requirements 73 a Students should consult the General Education section of the catalog for a list of approved courses in this category. b This course is approved for the Analyzing the Natural World General Education category. [c] Students who take CHEM 116 and CHEM 118 to fulfill the general chemistry requirement do not need to take CHEM 222. Instead they should enroll in one of the following: CHEM 314: Inorganic Chemistry; CHEM 452: Biochemistry; or CHEM 402: Chemical Information Systems and CHEM 444: Physical Chemistry III. ***Required in the College of Engineering*** **Courses** **Hours** ENGR 100—Orientation[a] 0[a] CHE 201—Introduction to Thermodynamics 3 CHE 205—Computational Methods in Chem. Eng. 3 CHE 210—Material and Energy Balances 4 CHE 301—Chemical Engineering: Thermodynamics 3 CHE 311—Transport Phenomena I 3 CHE 312—Transport Phenomena II 3 CHE 313—Transport Phenomena III 3 CHE 321—Chemical Reaction Engineering 3 CHE 341—Chemical Process Control 3 CHE 381—Chemical Engineering Laboratory I 2 CHE 382—Chemical Engineering Laboratory II 2 CHE 396—Senior Design I 4 CHE 397—Senior Design II 4 CHE 499—Professional Development Seminar 0 ECE 210—Electrical Circuit Analysis 3 CME 260—Properties of Materials 3 CS 109—C/C++ Programming for Engineers with MatLab 3 **Total Hours—Required in the College of Engineering 49**	***SAME***

a ENGR 100 is one-semester-hour course, but the hour does not count toward the total hours required for graduation. Technical Elective **Courses** — **Hours** ***One technical elective to be chosen from the following list of design-oriented courses:[a]*** — 3 CHE 392—Undergraduate Research (3)[b] CHE 410—Transport Phenomena (3) CHE 413—Introduction to Flow in Porous Media (3) CHE 421—Combustion Engineering (3) CHE 422—Biochemical Engineering (3) CHE 423—Catalytic Reaction Engineering (3) CHE 425 Nanotechnology Drug Deliver (3) CHE 431—Numerical Methods in Chemical Engineering (3) CHE 433 Process Simulation with Aspen Plus (3) CHE 438—Computational Molecular Modeling (3) CHE 440—Non-Newtonian Fluids (3) CHE 441—Computer Applications in Chemical Engineering (3) CHE 445—Mathematical Methods in Chemical Engineering (3) CHE 450—Air Pollution Engineering (4) CHE 456—Fundamentals and Design of Microelectronics Processes 930 CHE 494—Selected Topics in Chemical Engineering (3) **Total Hours—Technical Elective** — **3** a Possible technical elective credit for a 400-level CHE course not listed above will require departmental approval by petition to the Undergraduate Committee. b An appropriate design-related research project may be selected with the approval of the Department of Chemical Engineering.	**Courses** — **Hours** ***One technical elective to be chosen from the following list of design-oriented courses:[a]*** — 3 CHE 392—Undergraduate Research (3)[b] CHE 410—Transport Phenomena (3) CHE 413—Introduction to Flow in Porous Media (3) CHE 421—Combustion Engineering (3) CHE 422—Biochemical Engineering (3) CHE 423—Catalytic Reaction Engineering (3) CHE 425 Nanotechnology Drug Deliver (3) CHE 431—Numerical Methods in Chemical Engineering (3) CHE 433 Process Simulation with Aspen Plus (3) CHE 438—Computational Molecular Modeling (3) CHE 440—Non-Newtonian Fluids (3) CHE 441—Computer Applications in Chemical Engineering (3) CHE 445—Mathematical Methods in Chemical Engineering (3) CHE 450—Air Pollution Engineering (4) **CHE 451- Renewable Energy Technologies (3)** CHE 456—Fundamentals and Design of Microelectronics Processes 930 CHE 494—Selected Topics in Chemical Engineering (3) **Total Hours—Technical Elective** — **3** a Possible technical elective credit for a 400-level CHE course not listed above will require departmental approval by petition to the Undergraduate Committee. b An appropriate design-related research project may be selected with the approval of the Department of Chemical Engineering.
Courses — **Hours** Electives outside the CHE rubric — 3 **Total Hours—Electives outside the Major Rubric** — 3	SAME
BS in Chemical Engineering, Bioengineering Concentration Students in this concentration complete the following: **Required Courses [a]** **Technical Elective** CHE 422 — Biochemical Engineering — 3 **Electives**	SAME

Select two electives in nonmajor rubric category from the following:		5-7
BIOS 350	General Microbiology	
BIOS 351	Microbiology Laboratory	
CHEM 352	Introductory Biochemistry	
CHEM 452	Biochemistry I	
Total Hours		**8-10**

[a] *Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 130 semester hours for the degree.*

Sample Course Schedule

SAME

Freshman Year

First Semester	**Hours**
MATH 180—Calculus I	4
CHEM 122—General College Chemistry I (Lecture)	4
CHEM 123—General College Chemistry I (Lab)	41
OR CHEM 116 – Honors and Major General and Analytical Chemistry I	5
ENGL 160—Academic Writing I: Writing for Academic and Public Contexts	3
CS 109—C/C++ Programming for Engineers with MatLab	3
ENGR 100—Orientation[a]	1[a]
Total Hours	15

Second Semester	**Hours**
MATH 181—Calculus II	4
PHYS 141—General Physics I (Mechanics)	4
ENGL 161—Academic Writing II: Writing for Inquiry and Research	3
CHEM 124—General College Chemistry II (Lecture)	4
CHEM 125—General College Chemistry II (Lab)	1
OR CHEM 118– Honors and Major General and Analytical Chemistry II	5
Total Hours	16

SAME

Sophomore Year

First Semester	**Hours**
MATH 210—Calculus III	3
PHYS 142—General Physics II (Electricity and Magnetism)	4
CHEM 232—Organic Chemistry I	4
CHE 201—Introduction to Thermodynamics	3
General Education Core course	3
Total Hours	**17**

<table>
<tr>
<td>Second Semester Hours
MATH 220—Introduction to Differential Equations 3
CHEM 234—Organic Chemistry II 4
CHEM 233—Organic Chemistry Lab I 2
CHE 205—Computational Methods in Chem. Eng. 3
CHE 210—Material and Energy Balances 4
CME 260— Properties of Materials 3
Total Hours 19</td>
<td>SAME</td>
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<tr>
<td>Junior Year
First Semester Hours
ECE 210—Electrical Circuit Analysis 3
CHE 301—ChE Thermodynamics 3
CHE 311—Transport Phenomena I 3
CHEM 222—Analytical Chemistry 4
General Education Core course 3
Total Hours 16

Second Semester Hours
CHEM 342—Physical Chemistry I 3
CHE 312—Transport Phenomena II 3
CHE 313— Transport Phenomena III 3
CHE 321—Chemical Reaction Engineering 3
General Education Core course 3
Total Hours 15</td>
<td>SAME</td>
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<tr>
<td>Senior Year
First Semester Hours
CHE 381—Chemical Engineering Laboratory I 2
CHE 396—Senior Design I 4
CHEM 346—Physical Chemistry II 3
CHE Technical elective—Selected from CHE 410, 413, 421, 422, 423, 431, 438, 440, 441, 445, 450, 456, 494, or 392 (departmental approval is required for CHE 392) 3
General Education Core Course 3
Total Hours 15</td>
<td>Senior Year
First Semester Hours
CHE 381—Chemical Engineering Laboratory I 2
CHE 396—Senior Design I 4
CHEM 346—Physical Chemistry II 3
CHE Technical elective—Selected from CHE 410, 413, 421, 422, 423, 431, 438, 440, 441, 445, 450, 451, 456, 494, or 392 (departmental approval is required for CHE 392) 3
General Education Core Course 3
Total Hours 15</td>
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<tr>
<td>Second Semester Hours
CHE 382—Chemical Engineering Laboratory II 2
CHE 341—Chemical Process Control 3
CHE 397—Senior Design II 4
CHE 499—Professional Development Seminar 0
Elective outside the major rubric 3
General Education Core course 3
Total Hours 15</td>
<td>Same</td>
</tr>
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